

MODULE 2

Group

• denoted by $\{G, \cdot\}$

• set of elements with an operation

1) closure : if $a, b \in G$, then $(a \cdot b) = c \in G$

2) associative : $a \cdot (b \cdot c) = (a \cdot b) \cdot c \quad \forall a, b, c \in G$

3) identity element : if there is an element e in G such that $a \cdot e = a$
for addition, $e = 0$

multiplication, $e = 1$

4) inverse element : for each element a in G there is an element a' in G
$$a \cdot a' = e$$

* $\{-\infty, \dots, 0, \dots, \infty, +\}$ - Group

$\{0, 1, \dots, \infty, +\}$ - not a group

Abelian Group

A group that satisfies all 4 properties of a group & along with it, it satisfies commutative property.

$$a \cdot b = b \cdot a \quad \forall a, b \in G$$

* $\{a, b, c, d, \cdot\}$

operator

$\cdot$	a	b	c	d
a	a	b	c	d
b	b	c	d	a
c	c	d	a	b
d	d	a	b	c

$\Rightarrow$ this explains the operation

(1) closure is satisfied

$$a \cdot b = b \in G$$

$$b \cdot c = d \in G$$

$$a \cdot c = c \in G$$

$$c \cdot d = b \in G$$

$$a \cdot d = d \in G$$

$$b \cdot d = a \in G$$

(2) associativity satisfied

$$a(b \cdot c) = (a \cdot b) \cdot c$$

$$a \cdot d = b \cdot c$$

$$\underline{d} = d$$

(3) identity element $e = a$

$$a \cdot a = a \quad a \cdot b = b \quad a \cdot c = c \quad a \cdot d = d$$

(4) inverse element

$$a \cdot a = e = a$$

$\downarrow$
inverse of a

$$b \cdot d = a = e$$

$\downarrow$
inverse of b

$$c \cdot c = a = e$$

$\downarrow$
inverse of c

$$d \cdot b = a = e$$

$\downarrow$
inverse of d

(5) commutative satisfied

$$a \cdot b = b$$

$$b \cdot a = b$$

$$\underline{a \cdot b = b \cdot a}$$

Thus it is an abelian group

Cyclic Group

• every element is a power of a^k .

$$\{a^0, a^1, a^2, a^3, \dots, a^n\}$$

$\downarrow$
operator

$a \rightarrow$ generator

$$(1) a^0 = e \rightarrow \text{for } + a^0 = e = 0$$

$$(2) a^3 = a \cdot a \cdot a \quad \text{for } \times a^0 = e = 1$$

$$(3) a^{-3} = (a^1)^3$$

* Find generators of $G = \{Z_6, +\}$

$$Z_6 \rightarrow \{0, 1, 2, 3, 4, 5, +\}$$

$$0^0 \bmod 6 = 0$$

$$0^1 \bmod 6 = 0 \bmod 6 = 0$$

$$0^2 \bmod 6 = 0 \bmod 6 = 0$$

$$0^5 \bmod 6 = 0$$

$\downarrow$
operator is plus

$$1^0 \bmod 6 = 0$$

$$1^1 \bmod 6 = 1$$

$$1^2 \bmod 6 = 1 \bmod 6 = 1$$

$$1^3 \bmod 6 = (1+1+1) \bmod 6 = 3$$

$$1^4 \bmod 6 = 4$$

$$1^5 \bmod 6 = 5$$

$$2^0 \bmod 6 = 0$$

$$2^1 \bmod 6 = 2$$

$$2^2 \bmod 6 = 2 \bmod 6 + 2 \bmod 6 = 4$$

$$2^3 \bmod 6 = (2+2+2) \bmod 6 = 0$$

$$2^4 \bmod 6 = (2+2+2+2) \bmod 6 = 2$$

$$5^0 \bmod 6 = 0$$

$$5^1 \bmod 6 = 5$$

$$5^2 \bmod 6 = 10 \bmod 6 = 4$$

$$5^3 \bmod 6 = 15 \bmod 6 = 3$$

$$5^4 \bmod 6 = 20 \bmod 6 = 2$$

$$5^5 \bmod 6 = 25 \bmod 6 = 1$$

∴ Generators are 1 & 5.

Ring

- $\{R, +, \times\}$ - set of elements with 2 binary operations called addition and multiplication.
- Properties:

A1. Closure

A2. Associative

A3. Identity

A4. Inverse

A5. Commutative

M1. Closure

M2. Associative

M3. Distributive property : $a \times (b + c) = ab + ac$
 $(a + b) \times c = ac + bc \quad \} \forall a, b, c \in R$

→ If the ring satisfies commutative property for multiplication then it's called commutative ring.
ie it satisfies $a \times b = b \times a \quad \forall a, b \in R$

→ Integral Domain

~~The~~ This will satisfy multiplicative & non zero divisors property along with all above property.

Field (M5). Multiplicative identity

$$a \times 1 = 1 \times a = a \quad \forall a \in R$$

(M6) No zero divisors

If a and b is in R and $ab = 0$ then either $a = 0$ or $b = 0$

Fields

- $\{F, +, \times\}$ - a set of element with 2 binary operations called addition and multiplication.
- It follows axioms (A1) to (M6).
- It is an integral domain.

(M7) Multiplicative inverse

for each 'a' in F except 0 there is an element a^{-1} in F such that

$$a \times a^{-1} = a^{-1} \times a = 1.$$

* Find the ~~sub~~cyclic sub groups of Z_{10}^*

$Z_{10}^* \rightarrow$ set of numbers that are coprime to 10.

If a & b are coprime then $HCF(a, b) = 1$

$$Z_{10}^* = \{1, 3, 7, 9, \times\}$$

$$10 = 2 \times 5$$

$$\phi(10) = \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{5}\right) = 4$$

To find cyclic sub groups, stop the ~~inter~~ iteration when value repeats.

$1^0 \text{ mod } 10 = 1$ (since operator is multiplication)

$$1^1 \text{ mod } 10 = 1$$

$$1^2 \text{ mod } 10 = 1$$

$$3^0 \text{ mod } 10 = 1$$

$$3^4 \text{ mod } 10 = 81 \text{ mod } 10 = 1$$

$$3^1 \text{ mod } 10 = 3$$

$$3^2 \text{ mod } 10 = (3 \times 3) \text{ mod } 10 = 9$$

$$3^3 \text{ mod } 10 = 27 \text{ mod } 10 = 7$$

$$7^0 \bmod 10 = 1$$

$$7^1 \bmod 10 = 7$$

$$7^2 \bmod 10 = 49 \bmod 10 = 9$$

$$7^3 \bmod 10 = \frac{343}{343} \bmod 10 = 3$$

$$7^4 \bmod 10 = 2401 \bmod 10 = 1$$

$$9^0 \bmod 10 = 1$$

$$9^1 \bmod 10 = 9$$

$$9^2 \bmod 10 = 81 \bmod 10 = 1$$

$$9^3 \bmod 10 = 729 \bmod 10 = 9$$

$$9^4 \bmod 10 = 6561 \bmod 10 = 1$$

cyclic subgroups are: $\{1\}$

$$\{1, 3, 9, 7\}$$

$$\{1, 7, 9\}$$

$$\{1, 9\}$$

$$* -2 \bmod 6 = -2 + 6 = 4$$

$$-8 \bmod 6 = -2 + 6 = 4$$

↳ If b is additive inverse of a then $a+b=0$ $a+b=a+a^{-1}=e=0$

In case of modular arithmetic, b is the additive inverse of a if it satisfies

$$(a+b) \bmod n = 0$$

Eg: $(1+b) \bmod 6 = 0$ then $b=5$

↳ If b is multiplicative inverse of a then $a \times b = 1$

In case of modular arithmetic

$$(a \times b) \bmod n = 1$$

Eg: For mod 6, $(a \times b) \bmod 6 = 1$ 5x5

If we take 1 then $1 \times 1 \bmod 6 = 1$
 $5 \times 5 \bmod 6 = 1$

~~{1, 5}~~

→

$$\begin{aligned}(a+b) \bmod n &= (a \bmod n + b \bmod n) \bmod n \\(a-b) \bmod n &= (a \bmod n - b \bmod n) \bmod n \\(a \times b) \bmod n &= (a \bmod n \times b \bmod n) \bmod n\end{aligned}$$

$$\begin{aligned}11^7 \bmod 13 &= 11^{1+2+4} \bmod 13 = (11^1 \times 11^2 \times 11^4) \bmod 13 \\&= [11^1 \bmod 13 \times 11^2 \bmod 13 \times 11^4 \bmod 13] \bmod 13 \\&= \{11 \times 4 \times ([11^2 \bmod 13 \times 11^2 \bmod 13] \bmod 13)\} \bmod 13 \\&= \{11 \times 4 \times ([4 \times 4] \bmod 13)\} \bmod 13 \\&= (11 \times 4 \times 3) \bmod 13 \\&= 132 \bmod 13 \\&= \underline{\underline{2}}\end{aligned}$$

Modulo 8 Addition

+	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	0
2	2	3	4	5	6	7	0	1
3	3	4	5	6	7	0	1	2
4	4	5	6	7	0	1	2	3
5	5	6	7	0	1	2	3	4
6	6	7	0	1	2	3	4	5
7	7	0	1	2	3	4	5	6

To find additive inverse, find where the value is zero, then that column value is additive inverse of row value.

Modulo 8 Multiplication

x	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	2	4	4	8
3	0	3	6	1	4	7	2	5
4	0	4	0	4	0	4	0	4
5	0	5	2	7	4	1	6	3
6	0	6	4	2	0	6	4	2
7	0	7	6	5	4	3	2	1

In this table the elements ~~itself~~ {1, 3, 5, 7} is multiplicative inverse of itself. other elements don't have multiplicative inverse.

$$11^{220} \bmod 12 = (11^2 \times 11^4 \times 11^8 \times 11^{12} \times 11^{24} \times 11^{48} \times 11^{120}) \bmod 12$$

$$= 11^2 \bmod 12 \times 11^4 \bmod 12 \times 11^8 \bmod 12 \times 11^{12} \bmod 12 \times 11^{24} \bmod 12 \times 11^{48} \bmod 12 \times 11^{120} \bmod 12$$

$$= 11^2 \bmod 12 \times 11^4 \bmod 12 \times 11^8 \bmod 12 \times 11^{12} \bmod 12 \times 11^{24} \bmod 12 \times 11^{48} \bmod 12 \times 11^{120} \bmod 12$$

$$= 11^2 \bmod 12 \times 11^4 \bmod 12 \times 11^8 \bmod 12 \times 11^{12} \bmod 12 \times 11^{24} \bmod 12 \times 11^{48} \bmod 12 \times 11^{120} \bmod 12$$

$$11^{220} \bmod 12 = (11^{2+218}) \bmod 12$$

$$= (11^2 \bmod 12 \times 11^{218} \bmod 12) \bmod 12$$

$$= \{1 \times (11^2 \bmod 12 \times 11^{216} \bmod 12) \bmod 12\}$$

$$= \{1 \times (1 \times (11^2 \bmod 12 \times 11^{214} \bmod 12) \bmod 12) \bmod 12\}$$

$$= \{1 \times 1 \times 1 \times \dots \times 1\} \bmod 12$$

$$= 1$$

* Find multiplicative inverse of $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \bmod 11$

$$(1 \times 1) \bmod 11 = 1$$

$$(2 \times 6) \bmod 11 = 1$$

$$(3 \times 4) \bmod 11 = 1$$

$$(5 \times 9) \bmod 11 = 1$$

$$(6 \times 2) \bmod 11 = 1$$

$$(7 \times 8) \bmod 11 = 1$$

$$(8 \times 7) \bmod 11 = 1$$

$$(10 \times 10) \bmod 11 = 1$$

$$(1, 1), (2, 6), (3, 4), (5, 9), (7, 8), (10, 10)$$

$$-1 \bmod 12 = -1 + 12 = 11$$

Euclidean Algorithm

1) $\gcd(a, 0) = a$

2) $\gcd(a, b) = \gcd(b, a \bmod b)$

$r_1 \leftarrow a \quad r_2 \leftarrow b$

while $r_2 > 0$

{

$q = r_1 / r_2$

$r = r_1 - q r_2$

}

$r_1 \leftarrow r_2 \quad r_2 \leftarrow r$

$\gcd(25, 60)$

q	r_1	r_2	r
0	25	60	25
2	60	25	10
2	25	10	5
2	10	5	0
	5	0	

$\gcd(1970, 1066)$

q	r_1	r_2	r
0	1970	1066	904
1	1066	904	162
5	904	162	94
1	162	94	68
1	94	68	26
2	68	26	16
1	26	16	10
1	16	10	6
1	10	6	4
1	6	4	2
2	4	2	0
	2	0	

$\gcd = 2$

GCD(31245, 61456)

q	r ₁	r ₂	r
1	31245	61456	30211
2	30211 61456	30211	1034
29	30211	1034	225
4	1034	225	134
1	1034 ²²⁵ 134	134	134 43
1	134	43	5
2	43	5	3
1	5	3	2
1	3	2	1
2	2	1	0
	1	0	

gcd = 1

$$(d, 0) \text{ case} = (1 \times 1) + (0 \times 0)$$

Extended Euclidean Algorithm

• can be used to find multiplicative inverse.

• s_1, s_2 always initialize with 1 0

• $t_1, t_2 \rightarrow 0, 1$

$$r = r_1 - q r_2$$

$$s = s_1 - q s_2$$

$$t = t_1 - q t_2$$

q	r ₁	r ₂	r	s ₁	s ₂	s	t ₁	t ₂	t
	a	b		1	0		0	1	
	x	0							

Arrows indicate the flow of values from the previous row to the current row. Circles around the final values in the s and t columns indicate the result.

If the table is completed, then it satisfies,

$$(s \times a) + (t \times b) = \gcd(a, b)$$

* Suppose $a = 161$ $b = 28$

q	r ₁	r ₂	r	s ₁	s ₂	s	t ₁	t ₂	t
5	161	28	21	1	0	1	0	1	-5
1	28	21	7	0	1	-1	1	-5	6
3	21	7	0	1	-1	4	-5	6	-23
	7	0		-1	4		6	-23	

Arrows indicate the flow of values from the previous row to the current row. Circles around the final values in the s and t columns indicate the result.

$$sa + tb = (-1)161 + 6 \times 28 = 7 = \gcd(161, 28)$$

* Find the multiplicative inverse of 11 in $\mathbb{Z}_{26}$

$$r_1 = 26 \quad r_2 = 11$$

a in $\mathbb{Z}_n$

$$r_1 = n, \quad r_2 = a$$

while $r_2 > 0$

{

$$q = r_1 / r_2$$

$$r_3 = r_1 - q r_2$$

$$t = t_1 - q t_2$$

↓

update $r_2 = r_3$

$$r_1 = r_2$$

$$t_2 = t$$

$$t_1 = t_2$$

}

if $r_1 = 1$ then $a^{-1} = t_1$

q	r_1	r_2	r	t_1	t_2	t
2	26	11	4	0	1	-2
2	11	4	3	1	-2	5
1	4	3	1	-2	5	-7
3	3	1	0	5	-7	26
	1	0		-7	26	

$r_1 = 1$ that is, multiplicative inverse of 11 is $(-7 + 26) = \underline{\underline{19}}$

* Find multiplicative inverse of 12 in $\mathbb{Z}_{26}$

q	r_1	r_2	r	t_1	t_2	t
2	26	12	2	0	1	-2
6	12	2	0	1	-2	13
	2	0		-2	13	

Since $r_1 \neq 1$, 12 has no multiplicative inverse.

1* Find the multiplicative inverse of all elements in $\mathbb{Z}_{32}$

2) $7^{1024} \bmod 9$, $14^{323} \bmod 13$

3) GCD of (5035, 9092)

4) $3224x + 6024y = \text{GCD}(3224, 6024)$ find x & y

5) Find multiplicative inverse of 256 in $\mathbb{Z}_{3334}$ using Euclidean algorithm.

$$(2) 7^{1024} \bmod 9 = (7^{2+1022}) \bmod 9$$

$$= (7^2 \bmod 9 \times 7^{1022} \bmod 9) \bmod 9$$

$$= \{4 \times (7^2 \bmod 9 \times 7^{1020} \bmod 9) \bmod 9\} \bmod 9$$

$$= \{4 \times (4 \times 7^{1020} \bmod 9) \bmod 9\} \bmod 9$$

$$= \{4 \times [4 \times (7^2 \bmod 9 \times 7^{1018} \bmod 9) \bmod 9] \bmod 9\} \bmod 9$$

$$7^{1024} \bmod 9 = 7^{2+2+4+8+\dots}$$

$$= \{7^2 \bmod 9 \times \dots$$

$$7^{32} \bmod 9 \times$$

$$7^{512} \bmod 9\}$$

$$(7^2 \bmod 9 \times 7^2 \bmod 9) \bmod 9 = \{4 \times 4 \times [(4$$

$$= \{16 \times 7 \times [$$

$$= \{16 \times 7 \times 4 \times$$

$$= \{16 \times 7 \times 4 \times$$

$$= 9834496 \bmod$$

$$= \underline{\underline{1}}$$

$$14^{323} \bmod 13 = 14^{2+}$$

$$= \{14^2$$

$$14^3$$

$$14^4$$

$$= \{1 \times$$

$$= \{1 \times$$

$$= \underline{\underline{1}}$$

$$a^n \bmod (a-1) =$$

$$7^{1024} \bmod 9 = 7^{2+2+4+8+16+32+64+128+256+512} \bmod 9$$

$$= \{7^2 \bmod 9 \times 7^2 \bmod 9 \times 7^4 \bmod 9 \times 7^8 \bmod 9 \times 7^{16} \bmod 9 \times 7^{32} \bmod 9 \times 7^{64} \bmod 9 \times 7^{128} \bmod 9 \times 7^{256} \bmod 9 \times 7^{512} \bmod 9\} \bmod 9$$

$$(7^2 \bmod 9 \times 7^2 \bmod 9) \bmod 9 = \{4 \times 4 \times [(4 \times 4) \bmod 9] \times 7^8 \bmod 9 \times \dots\} \bmod 9$$

$$= \{16 \times 7 \times [(7 \times 7) \bmod 9] \times 7^{16} \bmod 9 \times \dots\} \bmod 9$$

$$= \{16 \times 7 \times 4 \times [(4 \times 4) \bmod 9] \times 7^{32} \bmod 9 \times \dots\} \bmod 9$$

$$= \{16 \times 7 \times 4 \times 7 \times 4 \times 7 \times 4 \times 7 \times 4\} \bmod 9$$

$$= 9832496 \bmod 9$$

$$= \underline{\underline{7}}$$

1092721

$$14^{323} \bmod 13 = 14^{2+4+8+16+32+64+128+64+4+1} \bmod 13$$

$$= \{14^2 \bmod 13 \times 14^4 \bmod 13 \times 14^8 \bmod 13 \times 14^{16} \bmod 13 \times$$

$$14^{32} \bmod 13 \times 14^{64} \bmod 13 \times 14^{128} \bmod 13 \times 14^{64} \bmod 13 \times$$

$$14^4 \bmod 13 \times 14 \bmod 13\} \bmod 13$$

$$= \{1 \times [(1 \times 1) \bmod 13] \times 14^8 \bmod 13 \times \dots\} \bmod 13$$

$$= \{1 \times 1 \times 1 \times \dots \times 1\} \bmod 13$$

$$= \underline{\underline{1}}$$

$$a^n \bmod (a-1) = 1$$

$$* 14x \equiv 12 \pmod{18}; x = ?$$

$$14x \pmod{18} = 12 \Rightarrow x = 6,$$

$$7x \pmod{9} = 6$$

If $a \pmod{b} \equiv c$ then $a \equiv c \pmod{b}$

$$14x \equiv 12 \pmod{18}$$

$$7x \equiv 6 \pmod{9}$$

$$x = (6 \times 7^{-1}) \pmod{9}$$

$$\cancel{7} \Rightarrow x \times 6 \pmod{9} \Rightarrow \cancel{7}^{-1} \Rightarrow (x \times \cancel{7}^{-1}) \pmod{9} = 1$$

$$\cancel{b=1} \quad \underline{\underline{b=4}}$$

4 is the inverse of 7.

$$x = (6 \times 4) \pmod{9} = \underline{\underline{6}}$$

$$* 10x \equiv 2 \pmod{15}$$

$$x = (2 \times 10^{-1}) \pmod{15}$$

$$10x \pmod{15} = 2$$

10 has no inverse in mod 15. $\therefore$ No solution.

Polynomial Arithmetic

- Field is of 2 types : Finite
Infinite

- Finite field

$$GF(p^n)$$

where p stands for prime.

$GF(5)$ is a finite field while $GF(6)$ is not ($\because 6$ is not prime)

$$GF(p) \equiv \mathbb{Z}_p$$

- Polynomial arithmetic has 3 types :

1. $3x^2 + 6x + 2$

2. coeff. are the numbers in $GF(p^n)$

ex: $GF(2)$ base $\{0, 1\}$ $x^6 + x^5 + 0x^4 + x^2$

3. $x^4 + 1 \text{ mod } x^3 + 2$

• $x^3 + x^2 + 2 \div x^2 - x + 1$

$$\begin{array}{r} x^2 - x + 1 \overline{) x^3 + x^2 + 2} \\ \underline{x^3 - x^2 + x} \\ 2x^2 - x + 2 \\ \underline{2x^2 - 2x + 2} \\ x \end{array}$$

$$\begin{array}{r} x^2 - x + 1 \overline{) x^3 + x^2 + 2} \\ \underline{x^3 - x^2 + x} \\ 2x^2 - x + 2 \\ \underline{2x^2 - 2x + 2} \\ x \end{array}$$

$$(x^2 - x + 1)(x^2 - x + 1) + x$$

* Multiply $x^7+x^5+x^4+x^3+x+1$ and x^3+x+1 in $GF(2)$

$$(x^7+x^5+x^4+x^3+x+1)(x^3+x+1)$$

$$= x^{10} + x^8 + x^7 + x^8 + x^6 + x^5 + x^4 + x^2 + x + x^7 + x^5 + x^4 + x^3 + x + 1$$

$$= x^{10} + 2x^8 + 2x^7 + 2x^6 + 3x^4 + 2x^3 + x^2 + 2x + 1$$

above is ordinary multiplication. To map to $GF(2)$, use mod 2

$$= \underline{x^{10} + x^4 + x^2 + 1}$$

$$2 \bmod 2 x^8 = 0x^8$$

$$3 \bmod 2 x^4 = x^4$$

*

$$\begin{array}{r} x^4+1 \\ x^3+x+1 \overline{) x^7+x^5+x^4+x^3+x+1} \\ \underline{x^7+x^5+x^4} \\ x^3+x+1 \\ \underline{x^3+x+1} \\ 0 \end{array}$$

• Irreducible polynomial

In $GF(2)$, $x, x+1$

$$x^2+x+1$$

$$x^3+x^2+1, x^3+x+1$$

$$x^4+x+1$$

} these are irreducible

* Is x^4+1 irreducible?

$$(x+1)(x^3+x^2+x+1) = x^4+x^3+x^3+x^2+x^2+x+x+1$$

$$= x^4 + 2x^3 + 2x^2 + 2x + 1$$

$$= x^4 + 1 \quad (\because \text{in } GF(2)) \therefore \text{it is reducible}$$

• $GF(2)$ has 2 elements $\{0, 1\}$

$GF(2^3)$ has 8 elements $\{000, \dots, 111\}$

• $x^2 \ x^1 \ x^0$

$$0 \ 0 \ 0 = 0$$

$$0 \ 0 \ 1 = 1$$

$$0 \ 1 \ 0 = x$$

$$0 \ 1 \ 1 = x + 1$$

$$1 \ 0 \ 0 = x^2$$

$$1 \ 0 \ 1 = x^2 + 1$$

$$1 \ 1 \ 0 = x^2 + x$$

$$1 \ 1 \ 1 = x^2 + x + 1$$

• $(x^2 + x + 1) + (x^2 + x + 1)$

$$= 2x^2 + 2x + 2$$

$$= 0 \text{ [since it is case of } GF(2^3)]$$

• $(x^2 + x + 1) \times (x^2 + x + 1)$

This gives a polynomial of x^3 . But $GF(2^3)$ has degree upto 2. So to convert it to $GF(2^3)$ range divide it with ^{a irreducible} ~~the~~ 3rd degree polynomial which is [irreducible polynomial] in $GF(2^3)$

$$(x^2 + x + 1)(x^2 + x + 1) = x^3 + x^2 + x^2 + x + x + 1$$

$$= x^3 + 1$$

$$x^3 + 1 \text{ mod } x^3 + x + 1$$

$$x^3 + x + 1 \begin{array}{r} 1 \\ x^3 + 1 \\ \hline x^3 + x + 1 \\ \hline 000 \\ \hline \end{array}$$

• $(x^5 + x^2 + x)(x^7 + x^4 + x^3 + x^2 + x)$
 Polynomial $x^8 + x^4 + x^3 + x + 1$

$$(x^5 + x^2 + x)(x^7 + x^4 + x^3 + x^2 + x)$$

$$= x^{12} + x^9 + x^8 + x^7 + x^6 + x^9 + x^6 + x^5 + x^4 + x^3 + x^8 + x^5 + x^4 + x^3 + x^2$$

$$= x^{12} + x^7 + x^2 \text{ mod } x^8 + x^4 + x^3 + x + 1$$

$$\begin{array}{r} x^4 - 1 \\ x^8 + x^4 + x^3 + x + 1 \overline{) x^{12} + x^7 + x^2} \\ \underline{x^{12} + x^8 + x^7 + x^5 + x^4} \\ -x^8 - x^5 - x^4 + x^2 \\ \underline{-x^8 - x^4 - x^3 - x - 1} \\ -x^5 + x^3 + x^2 + x + 1 \end{array}$$

$$= \underline{\underline{x^5 + x^3 + x^2 + x + 1}}$$

* In $GF(2^4)$. Find the multiplicative inverse of $x^2 + 1$

Irreducible polynomial of degree 4 = $x^4 + x + 1$

$t = t_1 \oplus 2t_2$ - in case of polynomial

$$\begin{array}{r} x^2 + 1 \overline{) x^4 + x + 1} \\ \underline{x^4 + x^2} \\ -x^2 + x + 1 \Rightarrow \end{array}$$

In each step convert to $GF(2)$ coeff.

$$-x^2 + x + 1 \Rightarrow \therefore -1 \text{ mod } 2 = 1 \Rightarrow x^2 + x + 1$$

q	r_1	r_2	r	t_1	t_2	t
x^2+1	x^4+x+1	x^2+1	x	0	1	x^2+1
x	x^2+1	x	1	1	x^2+1	$1+x^3+x$
x	x	1	0	x^2	$1+x^3+x$	x^4+x^2+x
	1	0				

$$t = t_1 \oplus q t_2$$

$$\begin{array}{r} x^2+1 \\ x^4+x+1 \\ \hline x^4+x^2 \\ \hline -x^2+x+1 \\ \hline -x^2-1 \\ \hline x+2 \end{array}$$

$$1+x^3+x \quad x^4+x^2+x \quad 1+x^3$$

since $r_1 = 1$, multiplicative inverse is $1+x^3+x$

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$